Electric Potential: The Calculus

Reading **E** off a contour map, finding *V* by integration, and getting every Gauss geometry for free

> WHERE THIS SITS
>
> **Note 4** did electric potential with algebra, and everything in it stands on its own. This note is the same subject with calculus in hand, and it is where the energy picture stops being merely tidier than the force picture and starts being *cheaper* — one scalar integral instead of two or three vector ones, and then a whole catalogue of fields turned into potentials for nothing.

Reading the field straight off the map

Here is the payoff for doing the scalar first: **if you know *V* everywhere, you already know E everywhere.** You never have to go back and do that vector integral at all.

Look again at the map of equipotential lines in **Figure 2 of Note 4**, and think of it as a contour map of a hill, because that is exactly what it is. Two things are true about **E** at every single point on that map:

- **E** points the way *V* falls — **straight downhill**.
- **E** is **perpendicular** to the equipotential line it is standing on.

How steep the hill is tells you how big **E** is:

$$E = \Delta V / \Delta r$$

the change in potential divided by the distance you walked to get that change. Where the lines are crowded together *V* is dropping fast, so the field is strong; where they are spread apart, the field is weak. **You can read the field strength straight off the spacing!**

That is all $\mathbf{E} = -\nabla V$ is saying. The minus sign is "downhill," and the gradient is "how steep." Notice that this lets you get the whole field off a contour map without doing a single integral, which on an exam is usually a good deal faster.

Going the other way — *V* by integration

We already know how to find the field of a charge that is spread out: you chop it into little pieces, find the d**E** from each piece, and add the pieces up. **The adding up is where it gets painful, though!** **E** is a vector, so "adding up" means components, and components mean two or three integrals with a lot of geometry to keep straight.

Now here is the good news. **The potential is much kinder, because *V* is a scalar.** Remember the two world's views from **Note 4**? This is exactly where the energy perspective earns its keep. You chop the charge into the very same little pieces, you write $dV = K\,dq/r$ for each one, and you add. **Just numbers, no directions, one integral.**

So it is the same recipe as before with the worst step crossed out:

1. Draw the charge, pick your little piece, say where it is.

2. Say where the point of interest is.
3. Write dq in terms of the charge density.
4. Write r, the distance from the piece to the point.
5. Integrate $K\,\mathrm{d}q/r$ over the whole charge.

And then, always, check it. Back away far enough and any finite blob of charge has to look like a point charge, so V should collapse to KQ/r. If it doesn't, it is your answer that is wrong, not the limit!

Charge	Point	Result	Far-field check
Finite rod, length L, charge Q	on its axis, distance x from the centre	$V = (KQ/L)\ \ln[(x + L/2)/(x - L/2)]$	for $x >> L$ this $\rightarrow KQ/x$ ✓
Quarter-circle arc, radius R, density λ	at the centre	$V = K\lambda\pi/2$	every piece is the same distance R away, so the integral is just the total charge over R
Half-circle arc, length L, charge $-Q$	at the centre	$V = -KQ\pi/L$	and the field there is **not** zero — $E = -2KQ\pi/L^2\,\hat{\imath}$

That last row is the lesson of this whole note in one line. **The arc has a perfectly good potential at its centre and a perfectly good field there too, and neither one tells you the other without doing the work.**

And the bridge to Gauss

In the Gauss note we find $E(r)$ for a whole catalogue of shapes — the solid insulating sphere, the metal sphere, the hollow shell, the coaxial cable. Each one takes a Gaussian surface, a symmetry argument and two lines of algebra.

We get the potential of every one of them for free. $\Delta V = -\int \mathbf{E}\cdot\mathrm{d}\mathbf{r}$, and the field is already in hand. All that is left is an integral along a radial line from infinity in to wherever you want V. Where E changes form — inside the sphere against outside it — integrate piecewise and match at the boundary: ***V* is continuous across the surface even where *E* jumps.**

The solid insulating sphere is the one to work carefully, because it shows both features at once. Outside, $V = KQ/r$. Inside,

$$V(r) = 3KQ/2R - KQr^2/2R^3$$

a parabola, flattening to its maximum at the centre. Graph it under $E(r)$ on the same axis and you can see the relationship: ***E* is the slope of *V***, and at the centre, where V is flat, E is zero.

Looking ahead

That is potential done twice, once each way. **Note 8** steps back from both and asks a different kind of question: why do the electrostatic equations look so much like Newton's, and what can you get for free once you notice it?